CIS 6270 Project 2: An ICML-Style Research Paper Section-by-Section Writing Guide and Illustrative Examples

Anonymous Authors[1]

Abstract

This guide contains writing instructions and illustrative examples. Write the final Project 2 abstract as one concise paragraph after completing the experiments. State the discrete generation problem, the limitation in prior methods, and the new generative mechanism. Identify the principal benchmark and comparison, report the strongest measured quantitative evidence with uncertainty where appropriate, and state the supported finding and its scope. Aim for four to six connected sentences, adjusting the length for technical completeness and readability. Use self-contained prose and concrete findings. *For the November 10 outline, describe the proposed formulation and planned evaluation, and label expected outcomes as hypotheses.*

How to use this template. This guide explains how to organize the **Project 2** paper, define mathematical notation, construct a method figure, format LaTeX floats and experimental tables, and connect claims to evidence. Replace the instructional paragraphs and placeholder tables with a paper about your method and measured results. Choose and justify the number of subsections, modalities, baselines, ablations, and independent seeds needed to support your claims. The official 40-point grading rubric appears in `Project2_Description.tex`.

Timeline and format. The research outline is due **November 10 at 8:30 am**; the complete paper and code are due **December 3 at 8:30 am**; defenses take place **December 4** (20-minute presentation and 10-minute Q&A). Use the anonymous *ICML main-conference submission* template. The ICML 2026 rule is provisionally *eight pages of main text*, excluding references and appendices, until the ICML 2027 instructions supersede it. This teaching guide expands each section for instruction; prepare the final paper within the assigned page limit. The main text must remain self-contained; required proofs, implementation details, and complementary results may appear in an appendix.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Build the Introduction around the problem, the unresolved technical gap, the proposed formulation, its expected mechanism, and the supporting evidence. Give each paragraph a clear role and connect it to the next. Choose the number of paragraphs to develop the research question and contribution.

Problem and motivation. Define the *discrete objects* the method generates: e.g., categorical sequences, graphs, discrete code tokens, molecular graphs, or variable-length objects. State the practical or theoretical capability that matters, such as sample quality, structure validity, likelihood, controllability, diversity, generation efficiency, or calibrated probability modeling. Discrete diffusion and flow methods are relevant examples of existing approaches (Austin et al., 2021; Campbell et al., 2022; Gat et al., 2024). Explain the task-specific motivation for the chosen discrete formulation and support comparisons with autoregressive methods or continuous relaxations using relevant evidence.

Technical gap and closest related method. Introduce the *closest* work and say what is already solved and what remains unresolved. For example, some papers study categorical corruption and reverse processes (Austin et al., 2021; Campbell et al., 2022); others introduce masked objectives (Sahoo et al., 2024), probability paths for discrete flow matching (Gat et al., 2024), discrete score-ratio estimation (Lou et al., 2024), or flows through the simplex (Stark et al., 2024; Davis et al., 2024). Select the references and baselines relevant to the topic. Your gap must be specific enough that a new formulation can plausibly address it. Describe the limitation in terms of the task, comparison conditions, and supporting evidence.

Proposed idea and research hypothesis. Describe the central mechanism in plain language before introducing

technical notation. State what element changes relative to the chosen base model (for example, the probability path, rate matrix, coupling, loss, state representation, objective, guidance, or sampler). Specify *what you expect to improve and why*. Explain the methodological difference in the Introduction and evaluate its effect in Results.

Evidence and contributions. Preview the experiments necessary to support the hypothesis: relevant benchmarks, competitive comparisons, controlled mechanism ablations, multi-seed evaluation, and any efficiency, sensitivity, or generalization analyses needed for your claim. In the final paper, summarize the main quantitative finding *accurately*, including trade-offs and negative results. Write contribution bullets after establishing the paper's findings.

Recommended contribution structure (adapt to your research):

- **Formulation:** We introduce [precise new mechanism] for [discrete generation task], addressing [concrete limitation].
- **Analysis:** We characterize [legality, assumption, objective, efficiency, or another supported property] under [stated conditions].
- **Evidence:** On [benchmarks], we measured [comparison and supported advantage or trade-off] and tested [central mechanism] through [controlled ablation].
- **Scope:** We tested [generalization, robustness, limitation, or failure condition] through [completed experiments].

Novelty statement (required content). Write a short explicit comparison of *closest prior work* $\rightarrow$ *components reused* $\rightarrow$ *specific new mechanism* $\rightarrow$ *why the difference matters*. Identify the contribution through its mathematical or algorithmic mechanism. Describe dataset, training, and hyperparameter choices as experimental settings. For a combination of published components, explain the new formulation and establish its technical difference from prior work.

Introduction review. Keep the background focused on the research question. Support novelty claims with a comparison to prior formulations, and describe contributions through the capabilities and findings established in the paper. Support statistical significance, performance rankings, and speed claims with the corresponding results.

2. Related Work

Use Related Work to position the new method technically. Organize the literature around the design choice the paper changes. Distinguish foundational methods, directly competing formulations, and work relevant to applications or evaluation. Cite each source for the specific result or method it establishes.

Table 1. Illustrative novelty-comparison layout. Replace each placeholder with a verified technical description of the compared methods.

Method	Shared basis	Key difference
Closest prior	[formulation]	[published mechanism]
Ours	[reused parts]	[actual new mechanism]

Foundational formulations. Choose the relevant families and explain the specific mathematical difference. Discrete-time categorical diffusion with structured transition matrices is exemplified by D3PM (Austin et al., 2021); continuous-time discrete denoising uses CTMCs and learned reverse dynamics (Campbell et al., 2022); discrete flow matching constructs generation through probability paths and associated updates (Gat et al., 2024). The continuous flow-matching viewpoint is developed by Lipman et al. (2023), and its continuous vector fields require an explicit connection to categorical jump dynamics when used for discrete generation. Discuss the families that the proposed method builds on or compares with.

Methods closest to the innovation. Discuss the actual mechanism in the most relevant competitors. Masked diffusion objectives and training recipes are studied by MDLM (Sahoo et al., 2024); score-based ratio modeling by SEDD (Lou et al., 2024); simplex transport choices by Dirichlet Flow Matching (Stark et al., 2024) and Fisher Flow Matching (Davis et al., 2024); structured categorical graph generation by DiGress (Vignac et al., 2023). Select baselines with matching generation tasks, data, and evaluation protocols.

Point-by-point difference from prior work. For each close paper, state: *its idea, the relevant assumption or limitation, what your method shares, and what is newly derived or designed*. The difference may lie in the transition law, path family, loss, training target, coupling, guidance target, decoder, solver, computational behavior, or structure constraints. For a change in neural backbone, explain how the change affects the generative formulation.

Use a comparison table when it helps explain the technical differences from prior work:

Citation integrity. Include complete entries for all cited work, preferably the peer-reviewed venue version when available. Check spelling, author order, year, title, venue, and identifier. Inspect each cited source and verify that it supports the stated theorem, method, or empirical result. *The supplied* `Project2_References.bib` *contains example references; extend it with sources relevant to the project.*

3. Preliminaries and Problem Setup

Introduce the concepts needed to understand the method, with consistent definitions, symbols, and conventions. Give readers enough information to reconstruct the learned distribution and the procedure for generating discrete objects.

3.1. Discrete Objects, Conditioning, and Evaluation Task

Define the discrete sample space $\mathcal{X}$ and its structure. For a fixed-length categorical sequence, one possible choice is $\mathcal{X} = \{1, \dots, K\}^L$, with vocabulary size K and length L. For variable-length sequences, graph objects, or edit operations, specify a different sample space and any validity or structural constraints. Describe the observed data $x \sim p_{\text{data}}$, conditioning variables c when applicable, and the desired conditional generator $p_\theta(x \mid c)$ or unconditional generator $p_\theta(x)$. Clarify training inputs, generated outputs, allowed constraints, and what qualifies as a valid sample.

If your method uses pretrained encoders, auxiliary predictors, a learned property reward, a tokenizer, continuous embeddings, or a discretization procedure, state which objects are fixed, trainable, or externally supplied. Specify what the learning and evaluation problem is. Define how datasets are separated to reserve test data for final evaluation.

3.2. Time Convention and Markov Dynamics

[Example only: Choose and adapt the equations that describe the formulation used by your algorithm.]

A source-to-data generation convention may use $t \in [0, 1]$, with $p_0 = p_{\text{base}}$ and $p_1 \approx p_{\text{data}}$. If the model is formulated as a *forward corruption* process from data to noise, define a different time parameter (or make the time reversal explicit) and maintain the same convention throughout Methods, code, and figures.

For a discrete-time chain on K categories, let the row-stochastic transition matrix Q_t satisfy $[Q_t]_{ij} = q(x_t = j \mid x_{t-1} = i)$. Then, using *row-vector* probability conventions,

$$q(x_t = j \mid x_0 = i) = [Q_1 \cdots Q_t]_{ij}. \tag{1}$$

This is one way to describe a categorical corruption model; its actual transition matrices, mask state, and schedule must be specified (Austin et al., 2021).

For a *continuous-time* Markov chain on a finite state space, define a row-convention generator R_t by

$$\begin{aligned} R_t(i, j) &\geq 0 \quad (i \neq j), \\ R_t(i, i) &= -\sum_{j \neq i} R_t(i, j). \end{aligned} \tag{2}$$

Writing the marginal p_t as a *row vector*, its master equation is

$$\frac{d}{dt} p_t = p_t R_t. \tag{3}$$

Enforce these sign and row-sum conditions when defining CTMC rates. For a sufficiently small h, $I + hR_t$ is a first-order approximation to the transition kernel (with h chosen to keep the approximation nonnegative). State the numerical sampling method implemented (Campbell et al., 2022).

3.3. Probability Paths, Couplings, and Continuous Relaxations

A discrete flow method may define a *conditional* probability path $q_t(x \mid x_0, x_1)$ between source and target endpoints, a coupling $\pi(x_0, x_1)$, and the marginal

$$p_t(x) = \sum_{x_0, x_1} \pi(x_0, x_1) q_t(x \mid x_0, x_1). \tag{4}$$

Explain the endpoints, how π is chosen or learned, how intermediate states are sampled, and how the process is realized by an applicable rate or transition mechanism (Gat et al., 2024). Specify the sampling mechanism that realizes the path q_t.

For a model that works on the probability simplex, specify $z \in \Delta^{K-1}$, the continuous path, objective, and decoding rule. For instance, the *illustrative* categorical decoder

$$x \mid z \sim \text{Categorical}(z) \tag{5}$$

requires a legitimate probability vector z. Argmax decoding is a different, deterministic rule with different distributional consequences. Dirichlet and Fisher flow-matching papers demonstrate distinct continuous geometric representations for discrete data (Stark et al., 2024; Davis et al., 2024). *Simplex or continuous-latent methods are allowed for Project 2 only when the new contribution substantively concerns discrete generation.*

3.4. Notation and Assumptions

Use a notation table when it helps readers follow the symbols. Define time endpoints, token and time indices, rate-matrix row/column convention, source/data states, condition c, and parameters θ, ψ. State whether transition independence, token factorization, reversibility, detailed balance, or smoothness are *assumptions*, *enforced properties*, or *proved conclusions*. Justify each claimed property through the construction, a proof, or an applicable cited result.

4. Proposed Method

Begin Methods with the intuition behind the proposed mechanism, then define the mechanism, training objective, inference algorithm, and computational costs. Introduce each symbol before using it and connect the equations through their mathematical relationships.

4.1. Method Overview and Core Idea

State the innovation in two to four sentences. Describe the inputs, computations, change to the base generative process, and outputs. Name the base method and identify reused components. Connect each block in Figure 1 to the definitions and equations in this section.

The TikZ example illustrates how to construct a vector method figure in LaTeX. Replace it with a diagram of the proposed mathematical mechanism, including the relevant states, dependencies, and training and sampling operations. Use flowchart elements where they clarify these relationships.

4.2. Novel Discrete Generative Formulation

Define the proposed probability path, jump rates, coupling, state space, decoder, training target, loss, guidance mechanism, or sampler. Explain the shared baseline structure, then define the new component and its mathematical relationship to that structure. Write equations for the formulation implemented in the study.

Illustrative rate-based mechanism. An illustrative construction is to use an estimated clean-data posterior $\hat{p}_\theta$ to build off-diagonal rates, e.g.

$$R_{\theta,\psi,t}(i, j \mid c) = \mathrm{softplus}\big(g_{\theta,\psi,t}(i, j, c)\big), \quad i \neq j, \quad (6)$$

with diagonal entries determined by Equation (2). The `softplus` function enforces *nonnegative off-diagonal values*. Establish transport of the intended marginal p_t through a derivation, and assess performance through experiments. Derive the relationship between the proposed path or rates and the intended probability evolution, and state the method-specific constraints. For a discrete kernel, geometric simplex flow, or jump-event sampler, give the corresponding equations.

Novelty test. Identify the mathematical or algorithmic change, its dependence on the discrete task, the reused components, the predicted effect, and the ablation that isolates that effect. Establish novelty through a technical comparison with the closest prior formulation.

4.3. Training Objective and Parameterization

Define every learned component, the distribution of training inputs, time sampling, supervision target, and the final objective. A general clean-token prediction loss *may* take the form

$$\mathcal{L}_{\mathrm{CE}}(\theta) = \mathbb{E}_{x_1,t,x_t}\big[-\log \hat{p}_\theta(x_1 \mid x_t, t, c)\big], \quad (7)$$

where the expectation uses $x_1 \sim p_{\mathrm{data}}$, $t \sim \nu$, and $x_t \sim q_t(\cdot \mid x_1)$. The conditional sampler q_t, time distribution ν, conditioning distribution, factorization of $\hat{p}_\theta$, and any weighting must be the *actual* ones. Choose the loss appropriate to the formulation. Some models optimize an ELBO or score/rate-specific objective; for example, MDLM and score-entropy models motivate different choices (Sahoo et al., 2024; Lou et al., 2024). Derive or cite the relevant training identity when it matters, and distinguish the theoretical target from the practical surrogate loss.

Describe the architectural details that affect the method. State trainable versus pretrained/frozen parameters, optimizer, schedule, batch size, checkpoint selection, hyperparameter search, and any parameter-count differences from the baselines. If training and inference use different procedures, define both.

4.4. Sampling Algorithm and Legality of Discrete Outputs

Write a sampling procedure that a reader could implement from the text. Define the initial state, discrete update distribution, number or schedule of updates, time discretization, correctors or refinements, stopping conditions, and how a valid final object is extracted. If your algorithm uses multi-site or block updates, clarify how simultaneous changes are sampled. If it uses variable-length objects, define insertions, deletions, and termination events.

For a CTMC or rate-based sampler, show how transition probabilities and waiting times are computed or approximated. State if you use a Gillespie-type event simulation, tau-leaping, Euler approximation, or a different algorithm. Report the validity constraints imposed on rates, time steps, probabilities, and the generated structure. When using a continuous relaxation, distinguish categorical sampling from argmax rounding and say which distribution is evaluated.

4.5. Theoretical Properties, Assumptions, and Cost

A new theorem is optional. Justify each central mathematical claim with a derivation or an applicable cited theorem under explicit assumptions. This applies to claims about valid jump rates, exact marginals, unbiased objectives, constraint preservation, reversibility, invariant distributions, convergence, and optimality. Label approximations and conjectures according to their status.

When efficiency is part of the claim, analyze the relevant scaling (for instance, vocabulary size, sequence length, number of transitions, model evaluations, or algorithmic memory). Distinguish theoretical complexity from measured wall-clock time and indicate whether preprocessing or external predictor calls are included. State the model's domain of applicability and label heuristics and approximations explicitly.

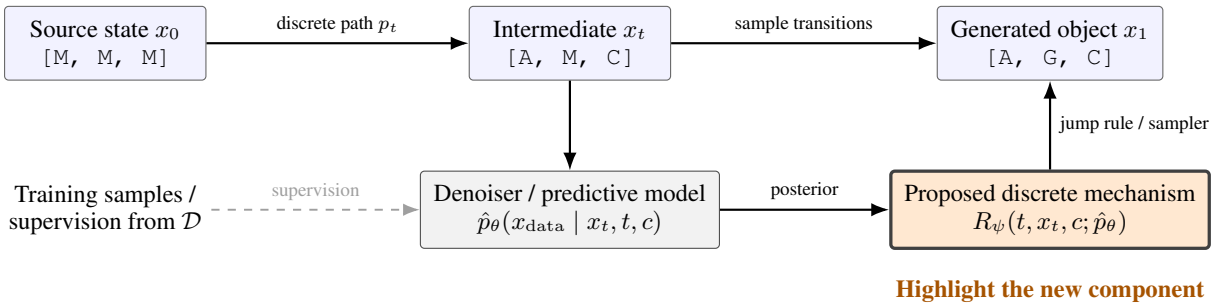


Figure 1. **Illustrative scientific method-overview schematic (replace before submission).** The top row shows example *discrete* states at different stages of generation; the lower row shows how a predictive model can inform a transition mechanism. The emphasized box represents where a proposed component could enter the mechanism. A final method figure must display the team's actual state space, training and sampling dependencies, exact novel component, and relevant constraints; its arrows, legends, symbols, and conditioning must match the proposed equations. Show the generative operations and their dependencies in the final figure.

4.6. Discrete Outputs from Continuous Latents

For methods using continuous representations, include this subsection. Show how a continuous simplex/latent representation connects to a target distribution over discrete objects and what your formulation contributes beyond an unchanged continuous generator plus decoder. Define the training representation, path, marginal, decoder $p(x \mid z)$ or deterministic map $D(z)$, and how discretization changes the outcome. When training uses a continuous surrogate and evaluation uses discrete samples, discuss the mismatch and provide validation of the final decoded distribution. Identify the new mechanism that connects the relaxation to discrete generation.

5. Experimental Setup

Describe the *experimental protocol* before presenting results so readers can identify the design decisions made before observing outcomes. A single-modality study can satisfy the requirements when its experiments support the claims. Choose and justify the number of benchmarks, baselines, independent seeds, and ablations according to the coverage and statistical power required.

5.1. Benchmarks, Splits, and Data Processing

For every benchmark, describe the discrete objects, task definition, data source, data version, number of examples, preprocessing, tokenizer/vocabulary if applicable, sequence or graph-size distribution, and train/validation/test construction. Explain duplicate filtering, target leakage, overlapping sequences, homologous families, or train/test contamination where relevant. If evaluation includes a different distribution or conditional design objective, define it explicitly. Report where the data can be obtained and which licenses or access restrictions apply.

5.2. Baselines and Fair Comparisons

Choose baselines for technical relevance and competitiveness. Explicitly cover the *unchanged base model*, *closest prior method*, and other *strong competitive approaches* as justified by the task; a method can satisfy more than one role. Appropriate starting references can include D3PM (Austin et al., 2021), continuous-time discrete denoising (Campbell et al., 2022), discrete flow matching (Gat et al., 2024), MDLM (Sahoo et al., 2024), SEDD (Lou et al., 2024), Dirichlet/Fisher flows (Stark et al., 2024; Davis et al., 2024), or task-specific graph methods (Vignac et al., 2023). *Select the methods appropriate to the task and justify the choice.*

For every comparison, make explicit which elements are held equal: dataset split, tokenization, conditioning, sample size, metric implementation, checkpoint selection, training budget, model size, optimization effort, and sampling budget. Disclose differences likely to affect the conclusion and account for them in the interpretation. If you copy literature-reported results, label them as such and explain any evaluation mismatch. For released pretrained checkpoints used when retraining is infeasible, report their provenance and training-data differences.

5.3. Metrics and Independent Validators

Choose metrics that test your research hypothesis. Task-specific examples include held-out log-likelihood/perplexity for language; validity, uniqueness, novelty, chemical feasibility, and graph property statistics for molecular graphs; diversity, distributional fit, motif recovery, and held-out property prediction for biological sequences; constraint satisfaction, functional test pass rates, or syntactic correctness for code; and generation cost (NFE, time, memory, query count) when efficiency matters.

Define each metric mathematically or cite a verified definition, its unit, the direction of improvement, aggregation,

Table 2. Illustrative claim–evidence map. Replace the placeholders with questions specific to your method before running the final experiments.

Claim	Controlled test	Evidence
Better quality	Strong baselines on same task	[primary metric]
Component helps	Full vs. matched control	[ablation metric]
More efficient	Equal quality / compute	[cost metric]

evaluation sample count, and limitations. For reward-guided methods, assess at least one independent quality, diversity, validity, or robustness metric alongside the optimized reward. If the training procedure uses an auxiliary evaluator, disclose predictor reuse and potential selection bias. Describe in-silico property scores as predictions; claims about physical activity, binding, or wet-lab outcomes require experimental measurements.

5.4. Training Budget, Randomness, and Statistical Reporting

Report parameter counts, compute/hardware, epochs or steps, learning rates, optimizer, batch size, checkpoint selection, training time, and generation budget when relevant. Explain which hyperparameters were tuned and on which validation set. State how many *independent* runs were performed, what was reseeded (model initialization, data order, training, sampler, evaluator), and how uncertainty was computed. Report samples from one checkpoint separately from independent model-training runs.

Define the use of mean $\pm$ standard deviation, standard error, bootstrapped intervals, confidence intervals, or paired tests; these are different quantities. Include sufficient numerical precision to compare methods at a precision supported by the measurements. Use multi-seed evaluation and justify its sample size and design for the task and computational cost. If baseline reruns are infeasible, describe that limitation and report the uncertainty estimates available for each source.

5.5. Controlled Ablations and Claim Mapping

Plan each ablation to test a specific component or explanatory hypothesis. Compare the proposed method with an appropriate base model or control. If there are several components, remove or replace them in a controlled way. Where relevant, match parameter count, number of evaluations, training data, or compute to isolate the effect of the proposed mechanism. An ablation can compare theoretically motivated alternatives for a mathematically inseparable component.

Table 3. **Illustrative main-results table with placeholders.** Report the mean $\pm$ standard deviation over the stated independent runs. Replace the illustrative Quality column with the study's metrics and define units and the aggregation protocol. Figure/table captions in the submitted paper must report the actual experimental setup.

Method	Quality ↑	NFE ↓
Closest prior	**TBD**	**TBD**
Strong baseline	**TBD**	**TBD**
Base model	**TBD**	**TBD**
Ours	**TBD**	**TBD**

6. Results and Analysis

Organize Results around the scientific questions tested. Present task performance, comparisons with the closest and strongest baselines, controlled tests of the proposed mechanism, and the observed trade-offs, costs, and failure conditions. Include cross-dataset or cross-modality tests when claiming generalization to those settings.

Every results subsection should state the question, comparison conditions, main observation, uncertainty and limitations, and a concise interpretation. Quantitative statements must agree with the tables, figures, abstract, appendix, and defense slides.

6.1. Main Quantitative Comparison

Report the complete main experiment on the proposed task, including all metrics needed to evaluate the core objective and major trade-offs. Include unfavorable metrics that affect the interpretation. State whether a gain is consistent across independent runs and whether an observed small difference is relative to uncertainty. Distinguish gains over the internal base model from gains over external methods.

Professional table requirements. Always use a `table` (or `table*`) floating environment, a descriptive `\caption` *above* the table, a `\label`, and `\toprule`, `\midrule`, `\bottomrule` from `booktabs`. Use open columns separated by `booktabs` horizontal rules. Align decimals appropriately; keep precision and measurement units consistent; explain the use of **bold** or underline for best results, any missing values, and the definition of error bars. Label placeholder tables as planned tables in the November 10 outline. In the final paper, replace every **TBD** with an actual result or an explained unavailable value.

6.2. Ablations: Does the New Mechanism Matter?

Use matched controls to isolate each proposed component and explain the observed differences. For a method with multiple novel elements, evaluate their separate effects through controlled removal or replacement. For an inseparable component, define a valid alternative mechanism and

Table 4. Illustrative controlled-ablation layout; all cells are placeholders. The actual paper must state what is held fixed between comparisons.

Configuration	Metric ↑	Cost ↓
Base model	**TBD**	**TBD**
Alternative mechanism	**TBD**	**TBD**
Full formulation	**TBD**	**TBD**

explain the comparison.

6.3. Sampling Efficiency, Calibration, or Trade-offs

Choose analyses that test the paper's claims. For an efficiency claim, report an interpretable cost measure (e.g., number of network evaluations, actual wall-clock latency, training budget, or property-predictor queries), together with generated quality. Assess speed improvements together with their effect on sample quality. For a guidance method, report the controlled property as well as preservation of validity and diversity. For a likelihood claim, use an appropriate held-out estimate with correct normalization and numerical precision.

If you use learning curves or quality–cost plots, label axes, units, legend, sample counts, and uncertainty clearly. Explain whether line shading reflects variation across independently trained models, repeated sampling from a fixed model, or uncertainty in a metric estimator, and identify which source of variation each interval measures. Show complete relevant ranges using axis ranges and selection rules that represent the observed behavior.

6.4. Robustness, Generalization, and Failure Modes

Examine the failure modes that matter to the new formulation: invalid discrete transitions, sharp sensitivity to a schedule or guidance strength, mode concentration, collapse in diversity, decoding mismatch, graph-constraint failures, instability at long sequences, or failures on out-of-distribution data when generalization is claimed. Include unfavorable examples and counterexamples when instructive, and make clear how samples were selected. State conclusions for the settings evaluated.

6.5. Numerical and Claim Consistency

For every stated percentage or absolute improvement, indicate the compared methods, dataset, metric, direction, and aggregation method. Ensure each reported number is identical when describing the *same setting* in the abstract, results text, tables, plots, appendix, and slides. A change of benchmark subset, seed aggregation, or evaluator must be labeled so readers can reconcile the reported values. Compute relative improvements from the actual stored results and check the formula, especially when a lower-is-better quantity is reported.

Interpreting negative results. Report positive, negative, and mixed outcomes and analyze possible explanations using controlled evidence. The rubric assesses original formulations and rigorous experiments under their respective criteria. Empirical-improvement credit requires convincing measured advantages. Include all tested conditions that affect the conclusions.

7. Discussion and Limitations

Interpret the main findings in relation to the proposed mechanism and the limitation identified in the Introduction. Explain the scientific implications of the measured results, distinguish observations from causal hypotheses, and state which explanations were directly tested.

Discuss major limitations that constrain interpretation: dataset size or leakage risk, untested scaling, compute constraints, mismatch in pretrained baseline data, proxy metric limitations, evaluator gaming, sensitivity to discretization, dependence on a single seed or checkpoint, or a theoretical guarantee that applies only in an idealized setting. A limitations paragraph should identify where the method may fail, why that matters, and what evidence would resolve the uncertainty. Support claims about biological function or deployment outcomes with direct evaluations in the relevant setting.

Discuss improvements, trade-offs, and negative or mixed findings in proportion to the evidence. Ground each experimental claim in the Results section.

8. Conclusion

Conclude with a short paragraph stating the discrete generative contribution, the strongest empirical evidence, the principal limitation or trade-off, and a next experiment motivated by those findings. Base the conclusion on the results and analysis presented in the paper.

Reproducibility and responsible disclosure. Follow the official ICML 2027 instructions when available for any required impact, AI-use, ethics, or additional statements. Regardless of venue policy, the paper and code must accurately acknowledge external models, datasets, evaluators, adapted figures, and substantial external code. Students are responsible for the correctness of every equation, reference, and result, including AI-assisted material. For anonymous submission, preserve anonymity in the title, affiliations, acknowledgments, metadata, and repository links.

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A. Appendix: Reproducibility Details

Place the central contribution, essential equations, competitive comparisons, principal ablations, and conclusions in the main paper. Use the appendix for expanded derivations, implementation details, secondary diagnostics, and supporting results, with references to the relevant main-text sections. Preserve anonymity where applicable.

A.1. Complete Dataset and Evaluation Details

Provide dataset versions, links, licenses when appropriate, access restrictions, train/validation/test splitting rules, filtering and de-duplication, data sizes, vocabulary/tokenization, graph construction or alignment, and each metric implementation including preprocessing, evaluator version, sample counts, and any confidence interval calculation.

A.2. Model Architecture, Training, and Sampling Configuration

Report architecture type, layer counts, hidden sizes, parameter counts, optimizer, schedule, weight decay, batch size, number of training steps/epochs, numerical precision, hardware, total compute, random seeds, checkpoint-selection criterion, and the actual sampling algorithm. Clearly identify external pretrained or frozen components and the exact datasets they saw when known. A concise configuration table often makes this information easier to verify.

Table 5. Illustrative reproducibility-details layout. Replace all placeholders with actual, method-specific configurations.

Item	Reported value
Architecture / parameters	**TBD**
Training dataset and split	**TBD**
Optimization	**TBD**
Time / noise / path schedule	**TBD**
Sampler / discrete update rule	**TBD**
Model and sampling seeds	**TBD**
Checkpoint selection	**TBD**
Hardware / runtime	**TBD**

A.3. Pseudocode for Training and Sampling

Use pseudocode when it improves reproducibility. Adapt the illustrative training and sampling algorithms below to the implemented intermediate-state construction, objective, and updates. Keep the source-to-target time convention consistent with Methods.

The paper's text, pseudocode, and code must all implement the *same* state space, transition law, training target, and sampling rule. Add an algorithm for a new coupling, adaptive schedule, guidance rule, or variable-length edit mechanism if central to the contribution.

Algorithm 1 Illustrative discrete generative model training (replace all generic steps)

Require: Training set $\mathcal{D}$, model parameters θ, method parameters ψ
Ensure: Selected parameters $\theta^\star$
for $k = 1, \ldots, K_{\text{train}}$ **do**
  Sample discrete data $x \sim \mathcal{D}$ and condition c if applicable
  Sample time t and intermediate state x_t using the declared path
  Construct the method-specific target or loss
  Update trainable parameters using the stated optimizer
end for
Select $\theta^\star$ using validation data
return $\theta^\star$

Algorithm 2 Illustrative discrete generation (replace with the implemented sampler)

Require: Trained model $\theta^\star$, condition c, sampling configuration
Ensure: Valid generated object x
Initialize $x_0 \sim p_{\text{base}}$ in the declared discrete state space
for each time interval / jump event in the declared schedule **do**
  Evaluate the denoiser, rates, or transition kernel as specified
  Sample a valid discrete update (and any method-specific corrector)
end for
Decode or validate the final object using the specified rule
return x

A.4. Additional Proofs and Correctness Checks

If your method claims valid normalization, legal CTMC dynamics, probability conservation, exact endpoint marginals, an unbiased objective, convergence, or a complexity result, provide the detailed proof under explicit assumptions. A direct *sanity test* of rate legality, a toy-domain comparison with known dynamics, a normalization check, or a deterministic regression test is valuable supporting evidence. If a proof is unavailable, state the property as a conjecture or narrow the claim to the available justification.

A.5. Extended Ablations, Per-Seed Data, and Failure Examples

Include experiments that test the choices (for example, schedule strength, sampling steps, initialization, coupling, guidance strength, or scaling), the results of additional seeds,

unsuccessful configurations that materially affect conclusions, and representative success and failure examples with explicit selection criteria. Distinguish repeated samples of one checkpoint from independently trained replicates. Additional baseline results should be labeled as recomputed, reimplemented, or sourced from the original papers.

A.6. Repository Structure and Reproduction Commands

Provide the precise installation instructions and commands needed to reproduce the main experimental claims and regenerate plots/tables. A reasonable repository structure might separate `configs/`, `data/`, `models/`, `train.py`, `sample.py`, `evaluate.py`, and `scripts/make_tables.py`; actual structure may differ. Include environment versions, hardware assumptions, checkpoints or access instructions, experiment configurations, result logs, and a small smoke test where feasible. If full-scale retraining is impractical for graders, provide verifiable stored results and a working demonstration of the core model logic.

B. Appendix: Presentation Checklist

B.1. Figures

A method figure must explain the actual mechanism; an experimental plot must test a stated claim. Use the `figure` or `figure*` environment; put *figure captions below the graphic*; use `\label` and `\ref` for cross-references; explain every panel, legend, variable, axis, direction, normalization, and uncertainty. Prefer vector PDF graphics or directly rendered diagrams; raster graphics must remain legible at final print size. Use accessible colors and distinguish curves with markers or line styles. Plot numerical results directly from the recorded outputs. Cite the source if an external figure is adapted.

B.2. Tables

Use `table` or `table*`; put *table captions above the table*; use `booktabs` horizontal rules with open columns; identify metric direction and units; align numeric precision; explain uncertainty, missing values, sample sizes, averaging, and highlighting. Keep every table entry consistent with the results text. Use labeled cross-references for tables and figures.

B.3. Equations, Language, and Citations

Define each symbol, give each symbol a consistent role, number and reference equations, and state whether a formula is derived, cited, or proposed. Connect each paragraph's definitions, operations, and conclusions explicitly. Use consistent terms for the same object and distinguish rates from probabilities, losses from likelihoods, and sampling draws from independent training replicates. Check every citation and every reproduced or adapted component. Use direct language and support each claim with the relevant mathematics or measurements.

B.4. Final Self-Check

Before submission, verify that (i) the paper proposes a substantive discrete-generative formulation and correctly identifies its closest prior work; (ii) all major equations and code agree; (iii) the main comparison is fair and uses adequate repeated evaluation; (iv) each ablation tests an explicit hypothesis; (v) all conclusions are supported, including negative cases; (vi) figure/table captions, legends, axes, and rules are professional; (vii) every numeric result matches across the manuscript, appendix, and slides; (viii) references have verified metadata; and (ix) the official ICML 2027 instructions have been checked for any updates to the provisional 2026 template. These checks summarize the criteria in the Project 2 description.